SINGAPORE DIMENSIONS • GRADE 3B • CHAPTER 9

Fractions: Equivalence & Common Denominators

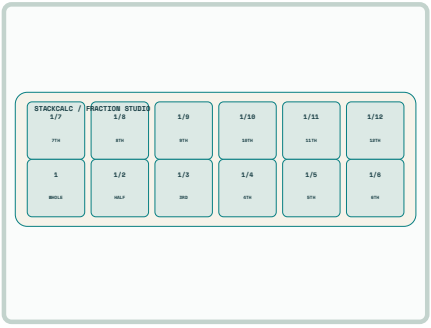
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.3.NF.A.3

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Tray

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Stack two 1/4 blocks (42mm) next to one 1/2 block (42mm). Observe equal heights.

Sketch Model on Paper:
Shade 1/2 of a circle and 2/4 of an equivalent circle.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	.	Step 1	Execute [.]
2.	2	Step 2	Execute [2]
3.	.	Step 3	Execute [.]
4.	4	Step 4	Execute [4]

EXPECTED LCD X: 1/2 (reduced by factor 2)

1 **Reduce 3/6 using the [SIMP] key.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Reduce 4/8 using the [SIMP] key.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Reduce 2/6 using the [SIMP] key.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Reduce 4/12 using the [SIMP] key.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Chapter 9

Fractions: Equivalence & Common Denominators • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Towers (84.0mm Whole)

CAD Blueprint: fraction_tray

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1 Bin & 1/1 to 1/12 Color-Coded Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Reduce 3/6 using the [SIMP] key.** Standard: CCSS.MATH.3.NF.A.3	1/2	[. 3 . 6] [SIMP]
2	**Reduce 4/8 using the [SIMP] key.** Standard: CCSS.MATH.3.NF.A.3	1/2	[. 4 . 8] [SIMP]
3	**Reduce 2/6 using the [SIMP] key.** Standard: CCSS.MATH.3.NF.A.3	1/3	[. 2 . 6] [SIMP]
4	**Reduce 4/12 using the [SIMP] key.** Standard: CCSS.MATH.3.NF.A.3	1/3	[. 4 . 12] [SIMP]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Towers must be seated firmly in the tray to compare vertical heights accurately.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.